Chapter 9

CORRELATION AND REGRESSION ANALYSIS

Learning Objectives

After studying this chapter students will be able to understand

- Concept of Karl Pearson's correlation co-efficient and the methods of computing it.
- Spearman's Rank correlation co-efficient
- Concept of Regression and Regression co-efficients
- Regression lines both x on y and y on x.

9.1 Correlation

Introduction


Karl Pearson

In the previous Chapter we have studied the characteristics of only one variable; example, marks, weights, heights, rainfalls, prices, ages, sales, etc. This type of analysis is called univariate analysis. Sometimes we may be interested to find if there is any relationship between the two variables under study. For example, the price of the commodity and its sale, height of a father and height of his son, price and demand, yield and rainfall, height and weight and so on. Thus the association of any two variables is known as correlation.

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Correlation is the statistical analysis which measures and analyses the degree or extent to which two variables fluctuate with reference to each other.

9.1.1 Meaning of Correlation

The term correlation refers to the degree of relationship between two or more variables. If a change in one variable effects a change in the other variable, the variables are said to be correlated.

9.1.2 Types of correlation

Correlation is classified into many types, but the important are

(i) Positive

(ii) Negative

Positive and negative correlation depends upon the direction of change of the variables.

Positive Correlation

If two variables tend to move together in the same direction that is, an increase in the value of one variable is accompanied by an increase in the value of the other variable; or a decrease in the value of one variable is accompanied by a decrease in the value of the other variable, then the correlation is called positive or direct correlation.

Example

(i) The heights and weights of individuals

(ii) Price and Supply

(iii) Rainfall and Yield of crops

(iv) The income and expenditure

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